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Developing research reasoning through digital tools in PhD students' methodological training

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Abstract

The paper examines how digital tools integrated into the course "Modern IT in Scientific Research" are associated with patterns of research reasoning among PhD students. The authors argue that the digital transformation of science affects not only the technical aspects of a researcher's work but may also shape how research problems are formulated, arguments are constructed, and evidence is verified. Research reasoning was operationalized through three indicators: I1—problematization, I2—methodological justification of procedures, and I3—reflective evaluation of evidence. These indicators are linked to mechanisms of digital tool use: M1—external representations and structuring of the research field (knowledge maps, citation graphs, timelines); M2—heuristic generation of alternatives and preliminary modeling; and M3—organization and verification of evidence (screening of contradictory citations, verification of claims, source management). The study employs a qualitatively dominant mixed-methods design. It draws on student work, reflective essays, educator observations, and a descriptive survey. The qualitative evidence suggests patterns of movement from procedural use of digital services toward more methodologically informed research reasoning. These patterns are reflected in students' attempts to refine research boundaries, justify methodological choices, critically evaluate findings, and detect errors produced by generative AI. At the same time, the study identifies common barriers, including technical difficulties, fragmented use of digital services without connection to research tasks, and limited reflection. A model for integrating digital tools into doctoral methodological training is proposed. The study contributes to understanding how digital technologies may be pedagogically integrated into the development of research reasoning in PhD programs.

Keywords PhD education, Digital tools, Artificial intelligence, Research reasoning, Digital literacy

1 Introduction

The digital transformation of science and higher education has changed the training of PhD students. It is no longer sufficient to simply master the tools. It is important to learn how to think through them: to ask precise questions, to construct arguments, and to verify claims. The widespread use of digital services (from bibliographic modeling platforms

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to generative AI systems) affects not only research techniques. It also influences the way ideas are generated, hypotheses are formulated, and knowledge is structured.

In this paper, we understand research reasoning as a process manifested through three indicators: I1 refers to problematization, which includes the precise formulation of a research question and the boundaries of the topic; I2 denotes methodological grounding, meaning the conscious choice of procedures, criteria for selecting sources, and methods of analysis; I3 involves the reflective evaluation of evidence, which includes checking the reliability of results and recognizing the limitations of methods and the risks of AI. We view digital tools as mechanisms that support the transitions: M1 includes external representations and the structuring of the research field, such as knowledge maps, citation graphs, and timelines; M2 refers to heuristic generation of alternatives and preliminary modeling; M3 involves the organization and verification of evidence, including screening for dissenting citations, verifying claims, and managing sources.

The focus of our study is on how these mechanisms specifically support indicators I1–I3 in the actual work of PhD students. This occurs during the course "Modern IT in Scientific Research". We present the Ukrainian case as an example of broader European shifts in digital preparation for research activities. The reference to the General Academic Competence Test (GACT) serves purely as contextual information. It functions as an analogue of GRE-like indicators of basic preparedness. It helps the reader understand the initial variation in students' cognitive and analytical skills. It is not used for modeling effects.

The research design is a qualitatively dominant mixed-methods approach. The core data consist of student artifacts, reflective mini-essays, and educators' observation journals. A descriptive survey ($n = 104$) is used as a contextual source and for triangulation with the qualitative findings, rather than as an explanatory or hypothesis-testing component. Data integration was organized through three strategies. First, merging involved comparing qualitative themes with descriptive survey findings. Second, joint displays aligned descriptive indicators with excerpts and analytical comments. Third, weaving embedded qualitative excerpts into the interpretation of descriptive trends.

The purpose of this study is to examine how digital tools, including AI-based ones, integrated into the course are associated with interpretative patterns of movement from procedural use toward research reasoning.

RQ1: Which mechanisms (M1–M3) are observed in students' outputs?

RQ2: How are these mechanisms related to indicators I1–I3?

RQ3: How do these observations align with the results of the survey?

Our contribution lies in several areas. First, we operationalize research reasoning (I1–I3). Second, we illustrate the roles of tools as mechanisms (M1–M3). Third, we demonstrate the integration of qualitative and descriptive data through joint displays. Finally, we propose reproducible pedagogical protocols for verifying evidence in PhD education.

2 Theoretical background

2.1 Literature review

The Vitae Researcher Development Framework [20] and the Open Science Skills Framework [15] provide normative guidelines for training future researchers. They emphasize personal effectiveness, knowledge, research skills, engagement, and scientific communication. They also highlight openness, transparency, reproducibility, data management,

and the ethical use of digital technologies. In this study, these frameworks are not reproduced descriptively. Instead, they serve as a "top-level" reference from which the operational core of research reasoning is derived. This core is then presented as a set of three indicators: I1—problematization, I2—methodological grounding, and I3—reflective verification of evidence. This reduction aligns with the idea that digital practices should be assessed not only as mastery of services, but also as the ability to construct and verify arguments in real research activities [12, 19].

Research indicates that digital tools influence the PhD students' reasoning process when they function as mechanisms rather than merely technical assistants. We identify three such mechanisms. M1 - external representations and structuring: knowledge maps, citation networks, and timelines transform simple "search" into problematization. They help refine research questions and define topic boundaries. They also shape the field of argumentation. M2—heuristic generation and preliminary modeling: generative systems, semantic graphs, and recommendation platforms accelerate the production of options and explanations. This effect appears only when explicit rules for source selection are applied and when inclusion and exclusion criteria are clearly documented [14, 16, 18]. M3—verification and organization of evidence: services such as Scite, Elicit, and Consensus support critical reading. They help identify disputing citations and organize bibliographic material. These functions directly strengthen reflective evaluation of the reliability of conclusions [1, 10]. From this perspective, digital tools operate as links in a chain "tool → procedure → evidence" rather than as isolated user skills.

Open science practices and data-management skills provide institutional conditions for M3. They ensure source transparency, traceability of changes, verification of claims, and repeatability of analytical steps [15]. Generative systems require ethical literacy. PhD students must understand the risks of fabrications, biases, and limits of applicability. This understanding must be combined with mandatory verification of claims and references [10, 19]. Recent findings also highlight a contradiction. Digital tools are widely used, yet pedagogical support remains fragmented. This gap leads to superficial practices that do not progress toward well-grounded argumentation [1]. To address this problem, instructional modules are needed. These modules should deliberately embed mechanisms M1–M3 into learning tasks with explicit criteria for evidence quality.

Standardized indicators of general academic skills, such as the GRE, are used in many systems to assess initial cognitive readiness. The Ukrainian TGAC (Test of General Academic Competencies) performs a similar function. In this study, it is mentioned only as a background signal of variability in students' initial analytical skills. It is not used for modeling effects. This framing makes it possible to relate the findings to broader European discussions on integrating digital literacy and epistemic competence in PhD education [9].

In this study, instead of providing a descriptive list of frameworks and services, we propose a synthetic foundation for further analysis. Research reasoning is a cognitive process that combines logical and analytical operations with methodological awareness and reflective evaluation of evidence. It differs from generalized academic or methodological competence. The focus is not on the sum of a learner's knowledge and skills, but on the way a well-grounded claim is constructed. This process spans from problematization to the selection of methods and the verification of the reliability of conclusions. In our model, research reasoning comprises three interrelated dimensions:

Analytical—the formulation of problems and working hypotheses, and the identification of logical connections;

Methodological—the conscious selection, combination, and justification of tools and procedures;

Reflective—the evaluation of the reliability of evidence, the limits of methods, and the ethical implications of using digital technologies.

The Vitae Researcher Development Framework describes the areas of researcher development, but it only partially addresses digital practices. The Open Science Skills framework specifies open science competencies, but it does not always capture the logic of moving from a digital tool to a well-founded conclusion. Our integration addresses these gaps. The analytical dimension combines Researcher Development Framework components on "knowledge and understanding of methods" with skills in open search and source evaluation. The methodological dimension links "research design and conduct" with open practices in data management, reproducibility, and transparency. The reflective dimension connects "professional standards and ethics" with open practices in citation, licensing, and evidence verification. Thus, the frameworks are not simply added together. They are woven into a sequence "tool → procedure → evidence → conclusion".

The proposed construct of research reasoning is related to, but not identical with, scientific thinking, methodological reasoning, and critical thinking. Scientific thinking is a broader construct that includes hypothesis generation, evidence evaluation, and the logic of explanation across different domains of inquiry. Methodological reasoning is more specific and concerns the selection, justification, and combination of research procedures. Critical thinking emphasizes evaluation, argument quality, and the identification of weaknesses in claims or evidence. Research reasoning, as conceptualized in this study, integrates these elements within the concrete process of doctoral research. Its specificity lies in the sequence from problem formulation to methodological justification and reflective verification of evidence in digitally mediated research activity. Therefore, the contribution of the I1–I3 model is not the introduction of a separate competence, but the operational description of how doctoral students construct, justify, and verify research claims when digital tools become part of their actual research workflow.

2.2 Operationalization of the construct "research reasoning"

The construct "research reasoning" is operationalized using three indicators, aligned with its dimensions.

I1. Problem formulation (analytical): clarity of the problem, working hypotheses, and relevant questions.

I2. Methodological justification (methodological): the appropriateness and combination of methods/tools for the task, and explanation of the choices made.

I3. Reflective evaluation (reflective): arguments regarding the reliability of evidence, limits of methods, and ethical considerations.

Operationalizing research reasoning through indicators I1–I3 allows it to be distinguished from general methodological competence. The focus shifts to the quality of evidence construction.

These indicators also make it possible to characterize levels of research reasoning.

Basic: the problem is formulated without a hypothesis (I1); a tool is named without explanation (I2); only declarative caution is expressed (I3).

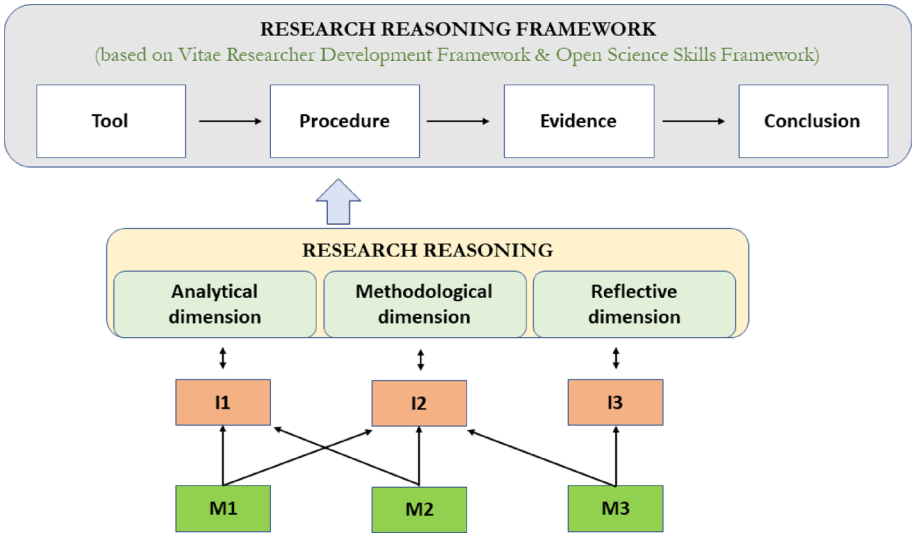


Fig. 1 Research reasoning model with operationalization (I1–I3) and mechanisms linking digital tools (M1–M3). Source: Own research

Table 1 Correspondence matrix

Tool/Service	Primary mechanism	Supported indicators	Typical evidence in work
Citation-based literature mapping tool (VOSviewer, ResearchRabbit, Connected Papers)	M1	I1, I2	Map/graph and justified selection of cluster or node
LLM (ChatGPT, DeepSeek, Claude)	M2	I1, I2	Drafts of formulation options with an explanation of edits
AI tools for literature review based on semantic search (Elicit, Scite, Consensus)	M2→M3	I1, I2, partially I3	Modification of the search strategy and a narrower corpus of sources
Bibliographic managers (Zotero + AI suggestions)	M3	I2, I3	Consistency of citations, notes, and tags for repeat verification

The examples of tools are provided to illustrate the mechanisms, not as exhaustive lists

Intermediate: the problem is formulated with 1–2 testable hypotheses (I1); the tool is linked to a procedure (I2); an example of source verification is provided (I3).

Advanced: the problem is broken into sub-questions with logical connections (I1); methods are combined with justification of sensitivity and limitations (I2); risks of errors and ethical implications are evaluated, and a method for verification is proposed (I3).

2.3 Mechanisms linking digital tools to research reasoning

The ability to use a digital tool cannot be reduced to a simple "tool = skill" equation. We describe three mechanisms through which tools influence reasoning (Fig. 1).

M1. External representation and knowledge structuring (visualizations, maps, networks) supports I1 and partially I2.

M2. Heuristic generation of ideas and solution options (language models, recommendation systems) supports I1 and I2, but requires subsequent verification.

M3. Evidence verification and source organization (citation screening, reference managers) directly strengthens I3 and regulates I2.

In this case, the corresponding matrix is shown in Table 1.

2.4 Design and implementation of the course

The course "Modern IT in Scientific Research", developed for doctoral students at Sumy State Pedagogical University named after A. S. Makarenko (Ukraine) and Kherson State University, implements the idea of moving from simple use of digital tools toward more methodologically informed research reasoning. In this approach, technologies function not only as supportive tools but also as cognitive mediators in the research process.

The course content is divided into three functional modules (Fig. 2).

The course objectives are built on a practice-oriented approach. Doctoral students complete personalized micro-projects focused on their dissertation topics. These include analyzing the research landscape using VOSviewer, constructing academic genealogies of researchers with the Mathematics Genealogy Project (for the specialty E7 Mathematics), classifying sources in Zotero, visualizing key concepts with Canva, etc. Special attention is given to the critical use of AI. Students learn to compare responses from different models, identify generation errors, and verify the validity of cited references.

The methodological approach to course design is based on the principles of open science, research-based learning, and digital pedagogy. The central pedagogical goal is not only to develop technical skills but also to create conditions for methodological responsibility regarding the outcomes generated through digital tools. The course includes reflective mini-essays, where doctoral students articulate their position regarding the limitations and potential of selected tools. They also analyze ethical challenges associated with AI use and reflect on how digital technologies may shape research reasoning practices.

3 Data and methodology

3.1 Research design

The research design is defined as a qualitatively dominant mixed-methods approach, incorporating an embedded single case of educational practice. This case involves the implementation of the course "Modern IT in Scientific Research" in the PhD programs

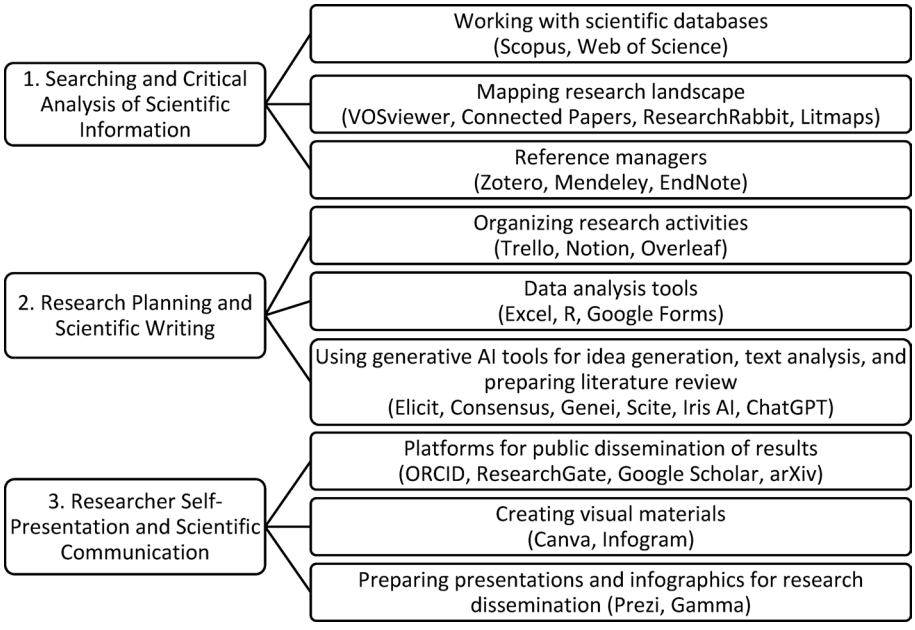


Fig. 2 The content of the course "Modern IT in Scientific Research". Source: Own research

at Sumy State Pedagogical University named after A. S. Makarenko and Kherson State University.

The qualitative component is dominant and includes a corpus of PhD students' papers and journals of pedagogical observations. Educator observations (5 individuals) were recorded as notes (Appendix B). The qualitative corpus comprised actual student cases, including reviews, maps in VOSviewer, working notes, and excerpts from project reports. It also included reflective mini-essays on working with AI and digital services, as well as survey results (Appendix A). The survey included open-ended questions about the reasons for selecting tools, difficulties encountered, and students' needs.

The main approach is reflexive thematic analysis, combining deduction and induction. An initial set of categories provided guidance for recording action functions (search, selection/filtering, critical reading/verification, synthesis/structuring, writing/formatting), cognitive levels of statements (description, explanation, justification, evaluation/reflection), and instrumental markers (LLM, citation-based literature mapping tool, AI tools for literature review based on semantic search, bibliographic managers, etc.). During the analysis, themes were refined using constant comparison and consideration of boundary cases. Analytical notes and a decision journal were maintained throughout the process.

The quantitative component was based on a semi-structured survey of 104 participants and documented common practices in the use of digital tools. The qualitative and quantitative data were collected concurrently over two academic years, analyzed separately, and subsequently integrated. Survey findings were reported as proportions of tool use and average ease-of-use ratings. The survey served a contextual and triangulation function rather than an explanatory one: it was used to characterize common digital practices and to examine their convergence with, or divergence from, the qualitative themes. The quantitative data were not used to test relationships, explain outcomes, measure change in research reasoning, or support causal inferences.

Data integration strategies [5] included:

Merging: comparing thematic analysis findings with descriptive survey indicators; correspondences and discrepancies were recorded in joint displays and brief analytical notes;

Joint displays: aligning qualitative themes with descriptive indicators;

Weaving: integrating qualitative excerpts into the interpretation of quantitative trends to show how tools specifically support problem formulation, procedural justification, and reflective evaluation of their use;

Triangulation: using educator observations to verify the stability of conclusions drawn from reflections and open-ended responses, and to refine interpretations in cases of discrepancy.

A combination of researcher reflexive analysis, peer debriefing in a small group, negotiated agreement on the interpretation of codes and themes, member checking of selected interpretations, and examination of negative cases was applied. The approach did not involve counting theme frequencies or calculating inter-coder coefficients. The focus was on aligning evidence across sources and transparently documenting decisions.

The operationalization of "research reasoning" used three indicators: I1—formulation of problems and questions; I2—methodological justification of procedures and selection criteria; I3—reflective evaluation of the reliability of evidence and the limits of AI use.

Interaction with tools is described through mechanisms M1 (external representations and structuring), M2 (heuristic generation of options), and M3 (evidence verification and source organization). These markers were specifically used to interpret the data in the Findings section.

The object of study is the educational course as a holistic phenomenon. Therefore, the research did not aim to calculate quantitative gains in indicators. The study focused on how PhD students' cognitive and methodological reasoning strategies were manifested in their use of digital tools during course assignments. The conclusions are presented as analytical generalizations.

3.2 Participants

The study involved 104 PhD students from Sumy State Pedagogical University named after A. S. Makarenko and Kherson State University, who took the course "Modern IT in Scientific Research" over two academic years. The qualitative corpus included the work of students who provided informed consent for the use of their learning materials for research purposes and agreed to depersonalization. The participants represented a diverse range of specialties and stages in their PhD programs. This diversity made it possible to consider variation in students' modes of digital interaction.

The data sources included:

Educational process papers: excerpts from literature reviews, knowledge maps and citation networks, working notes on search strategies, project reports, and elements of scientific writing;

Reflective mini-essays on experiences using AI and digital services during students' own research;

Semi-structured survey ($n = 104$) with closed and open-ended questions on tool usage practices, ease-of-use ratings, and difficulties;

Educator observation journals documenting typical action patterns, challenges, and strategies to overcome them during the course;

Course reference materials (syllabus, assignments, assessment criteria) necessary for interpreting the materials.

3.3 Data corpus and analysis protocol

The data corpus consists of textual products from the educational process: reflective essays, drafts of literature reviews, working notes on search strategies, and excerpts from project reports. The unit of analysis is a meaningful segment (a sentence or a short paragraph of up to ~120 words) containing a complete statement about an action, tool, or evaluation. Initial segmentation was performed before coding to align units across different types of materials.

Only materials created during the completion of course assignments and containing explicit signs of research activity were included in the analysis. These activities include searching, selecting and filtering sources, critical reading, synthesizing conclusions, scientific writing, and citation formatting. Technical files without substantive content, templates without original text, and materials containing personal data that could not be properly depersonalized were excluded.

Before processing, all documents were depersonalized: names, contacts, and other identifiers were removed. Each participant was assigned a code in the format R-__.

Agreed versions of the materials were stored separately from the coding tables. The use of learning products for research purposes was conducted with informed consent and did not affect the students' assessments.

Coding was conducted using a combined scheme that integrates functional and cognitive features with instrumental markers. Functional codes included F1—source search, F2—selection/filtering, F3—critical reading/verification, F4—synthesis/structuring, and F5—scientific writing/formatting. Cognitive codes included C1—description, C2—explanation, C3—justification, and C4—evaluation/reflection. Instrumental codes included T1—citation-based literature mapping tools (VOSviewer, ResearchRabbit, Connected Papers), T2—LLM assistants (ChatGPT, DeepSeek, Claude), T3—AI tools for literature review based on semantic search (Elicit, Scite, Consensus), and T4—bibliographic managers (Zotero, Mendeley). For each fragment, one functional and one cognitive code were assigned, and an instrumental code was added when a specific service was explicitly mentioned.

The analytical procedure followed a reflexive thematic approach. Initially, the corpus was coded with the recording of analytical memos. Themes within the codes were then refined using constant comparison and examination of boundary cases. Interpretations were aligned through analysis by two researchers, followed by a negotiated agreement and maintenance of a decision log. When needed, discussions were conducted within the research group (peer debriefing) and selected interpretations were clarified with students (member checking).

Evidence was integrated using joint displays, where mentions of tools in open-ended survey responses were compared with thematic fragments from students' work and educator observations.

Confidence in the findings was strengthened through source triangulation, analytical reflexivity, discussion of interim results within the team, documentation of negative cases, and careful recording of decisions. We did not calculate theme frequencies or conduct statistical tests of effects, instead prioritizing in-depth description of patterns and their alignment across sources.

The excerpts presented in the Findings were selected through analytical sampling. They were not intended to represent the most frequent responses. Instead, they illustrate analytically important patterns corresponding to mechanisms M1–M3 and indicators I1–I3. The excerpts were selected as clear instances of the identified patterns, while negative and boundary cases were considered during theme refinement.

In summary, the analytical workflow consisted of four sequential stages:

(1) Preparation and segmentation of the corpus;
(2) Coding using functional, cognitive, and instrumental markers;
(3) Thematic interpretation through mechanisms M1–M3 and indicators I1–I3;
(4) Integration with descriptive survey findings through triangulation and joint displays.

3.4 Ethical considerations

The study was approved by the Ethics Committee of Sumy State Pedagogical University named after A. S. Makarenko (Protocol No. 3, 2024).

Participants were informed about the research objectives, the types of data collected, and how these data would be used. Students provided consent for the analysis of their

learning materials, and participation did not affect their assessments. Personal data were removed before analysis, and quotations are presented in depersonalized form (R-).

The main risks were related to the possibility of identifying an author based on the content of a case or the topic of a dissertation. To mitigate this risk, topic titles were generalized, minor details were edited, and rare markers were removed. All references to third parties that could indicate a specific participant were eliminated.

No tools requiring the upload of personal materials to open services were used during analysis. In PhD student work where AI was used, potential source fabrications and concept mixing were checked during interpretation. For key examples, verification against primary publications was required.

Data were stored in secure folders with restricted access. Participant identifiers and coding tables were kept separately. Working files were versioned, and the decision log recorded dates, authors of changes, and brief explanations. Only aggregated results and anonymized quotations are presented in the published article. The underlying anonymized research data are available from the corresponding author upon reasonable request, subject to ethical and institutional requirements.

The study relies on educational artifacts and therefore does not measure skill gains through external tests, nor does it interfere with standard forms of assessment. Generalizations should be made cautiously; involving larger samples and additional indicators could enhance reproducibility in future studies.

4 Findings and discussion

4.1 From procedural use of digital tools to strategic reasoning

In the analyzed course materials, some fragments reflected basic activities, such as compiling bibliographies, creating simple visualizations, or preparing presentations. Other materials showed more purposeful uses of tools in relation to students' own research tasks. These included structuring the logic of literature reviews, visualizing the structure of a research field, and tracing trends or contradictions in current studies. The following analytically selected examples illustrate patterns consistent with movement from procedural tool use toward more methodologically informed reasoning.

Box 01 Evidence Verification and Critical Re-evaluation (M3 → I3).

> R-43, F3/C4/T1: "*Before citing an influential article, I used ResearchRabbit to verify it. The tool showed that although the article had 100 + citations, 15 of them were contradictory and pointed out methodological flaws. This made me critically reassess its reliability. In my work, I cited the article but also noted the existing criticisms and limitations identified using AI.*"
>
> R-101, F3/C4/T1: "*After ConnectedPapers showed that the key article I relied on had 12 contradictory citations, I reflected on the reliability of my entire theoretical framework. I analyzed the arguments in these refuting articles and decided to revise the introduction by adding a section on scientific debates surrounding this theory. This allowed me to present a more nuanced and critical perspective.*"

Analytical comment: both excerpts illustrate reasoning aligned with I3. Mechanism M3 guides disciplined work with evidence.

Box 02 LLM Heuristics and Defining Boundaries (M2 → I3).

> R-21, F4/C4/T2: "*I used ChatGPT to justify the boundaries of my research. I presented my preliminary structure to it and asked it to identify potential weaknesses in my argumentation. The AI suggested three alternative perspectives, two of which I integrated into the "Research Limitations" section, thereby enhancing the academic integrity and objectivity of my work*."
> R-55, F4/C4/T2: "*I used Claude to evaluate my own reasoning. After writing the conclusion, I uploaded the text to the AI with the prompt: 'Assess the strength of my arguments and identify possible logical errors.' Receiving the response, I realized that one of my arguments was circular. I rewrote it, which significantly improved the quality of the final conclusion. This was critical self-reflection initiated by AI*."

Analytical comment: LLMs act as a trigger for reflection, but the conclusions are formulated by the student.

Box 03 Selection and Transparency of Procedures (M2→M3 → I1/I2).

> R-32, F2/C2/T3: "*Elicit found 120 articles for me using a keyword. I used its filtering functions by publication year (last 5 years), methodology (only clinical studies), and availability of full text. This narrowed the list to 25 key works, which I then analyzed. Without this, I would have spent weeks reviewing unnecessary material*."
> R-46, F3/C4/T2: "*ChatGPT provided me with a list of sources on a given topic. Instead of accepting them as accurate, I critically reviewed the list: I checked whether these articles exist in Google Scholar, whether the authors are real researchers, and whether the AI had 'invented' them. This allowed me to filter out false sources and find genuine, relevant works*."

Analytical comment: field refinement (I1) is combined with transparent selection criteria (I2).

In the students' materials, a recurring pattern concerned practices of evidence verification and critical re-evaluation of the theoretical foundations of scholarly inquiry. A typical pattern combined F3 (verification) with C4 (reflection) and the instrumental marker T1 (citation-based literature mapping tool). For example, R-43: "Before citing an influential article… the tool showed… 15 contradictory citations… critically reassess its reliability…"; R-101: "…12 contradictory citations… revised the introduction by adding a section on scientific debates…". Such fragments directly supported indicator I3 (reflective evaluation) and demonstrate how mechanism M3 structures work with evidence. In some cases, verification also supported I2 (methodological justification), as in R-11, where a digital tool helped justify the choice of methodology based on aggregated evidence.

LLM tools were primarily used as heuristics for generating options and structuring material (M2). However, the work clearly documents reflective delineation of their limits. R-21: "*…used ChatGPT to justify the boundaries of my research… added two alternative perspectives in the "Limitations' section*" (F4, C4, T2 → I3). R-55 demonstrates self-reflection as a quality control of the conclusion: "*…identified a circular argument… rewrote it…*" (F4, C4, T2 → I3). At the same time, LLMs assisted at early stages of structuring: R-83 *("…Jenni.ai suggested the logic for the 'Discussion' section*") supporting I1→I2 without implying tool autonomy.

Table 2 Alignment of mechanisms and indicators

Mechanism	Indicator	Data	Comment
M3 Evidence verification	I3 Reflective evaluation	Box 01 (R-43, R-101)	Citation-based literature mapping tool → identification of contradictory citations → revision of position
M2 LLM heuristics	I3 Reflective evaluation	Box 02 (R-21, R-55)	LLM serves as a trigger, conclusions formulated by the student
M2→M3 Selection	I1, I2	Box 03 (R-32, R-46)	AI tools for literature review based on semantic search → filtering the corpus, source verification

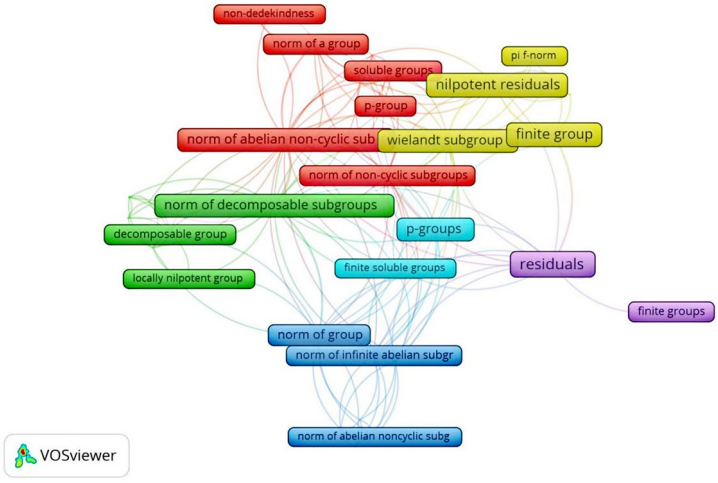


Fig. 3 Map of Keywords in VOSviewer.

The use of Elicit illustrates substantial narrowing of source "noise" and explicit formulation of selection criteria. R-32: "…120 articles → filters by year/methodology/full text → 25 key works…" (F2, C2, T3). Here, M2 (heuristics/filters) transitions into M3 (organization/screening), supporting I1 (refinement of the problem field) and I2 (transparency of selection procedures). R-46 reinforces this approach through verification of AI-suggested sources ("…*filtered out false sources…*"), this is M3 → I3.

The approach we applied to align mechanisms and indicators is summarized in the Table 2.

Visualization tools for literature review were used when it was necessary to clarify key concepts and outline topic clusters. This is evident in the maps created in VOSviewer, which highlighted central concepts and their connections (Fig. 3). A typical illustration from a student's note: "*After the initial mapping, I changed the set of keywords and obtained a separate cluster for Topic X, which clarified the boundaries of the review*" (R-26).

For citation network analysis, actions focused on selecting and structuring sources. ResearchRabbit supported the identification of relevant publications based on a seed paper and helped reorganize the field logic by reviewing nodes and temporal trajectories (Fig. 4). Connected Papers was used to find related papers through co-citation and bibliographic coupling when it was necessary to formulate hypotheses or identify interdisciplinary intersections (Fig. 5). An example of reasoning: "*The graph of related papers showed a node with methodological articles I had previously overlooked, which changed the structure of the section*" (R-38).

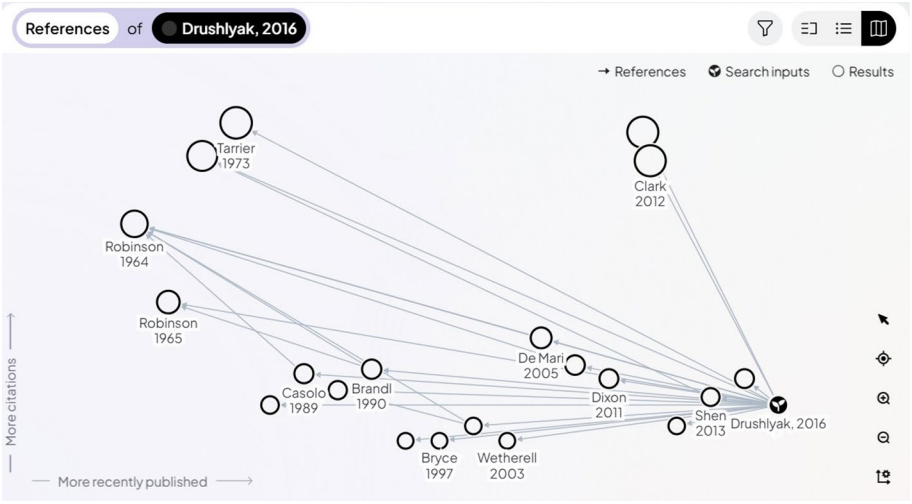


Fig. 4 Network in ResearchRabbit (based on [3]).

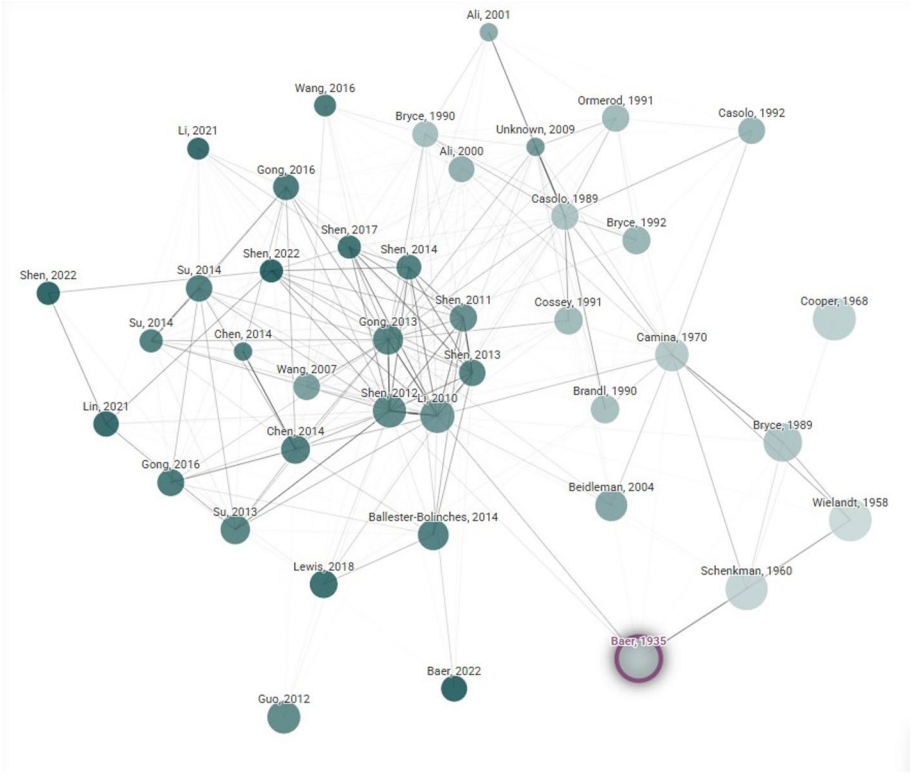


Fig. 5 Network in connected papers (based on [3]).

The use of AI tools was confirmed through content analysis of open-ended responses. The share of respondents who mentioned these services was as follows: LLM – 58 participants (56%), AI tools for literature review based on semantic search – 34 (33%), bibliographic managers – 23 (22%), citation-based literature mapping tool – 7 (7%).

In the responses, Elicit was most often associated with building search strategies for literature reviews and narrowing selections ("Accessibility," R-65; "Speed," R-76). ChatGPT was linked to rapid generation of drafts and formulation options ("Convenience

and ease," R-8; "Accessibility and speed of draft generation," R-70). Scite was associated with critical reading and verification of claims ("Simplifies work and analysis," R-44; "Saves time on verification," R-91). Zotero with AI suggestions supported organization of sources and notes ("Organization of sources," R-22; "Speed of note-taking," R-82).

At the same time, students consistently included reflective evaluations of AI limitations. They distinguished the convenience of automated suggestions from the need to verify conclusions using full texts, rejecting blind automation in favor of reasoned decisions. A typical statement was: "*The generated response provided a search direction, but I make the final conclusions only after reading the full texts*" (R-6).

4.2 Enhancing reflection on the research process

In the students' materials, a recurring pattern can be seen in the movement from accepting AI suggestions "as is" toward reflexive verification and correction of students' own decisions. A typical pattern combines F3/F4 (verification, synthesis) with C4 (reflection) and instrumental markers T1–T3 (Connected Papers, LLM, Elicit). These operate through mechanisms M2 (heuristics) and M3 (evidence checking) on indicators I3 (reflective evaluation) and I2 (methodological justification).

Students use LLMs as a trigger for self-checking conclusions, not as a replacement for analytical work. For instance, in R-21, ChatGPT helped identify "blind spots" and formulate research limitations; in R-55, the AI initiated the detection of a circular argument, after which the conclusion was rewritten. This illustrates M2 → I3: a heuristic prompt triggers reflexive revision of the argumentation.

Box 04 Illustrative excerpts (reflection, verification, selection).

> Materials R-21 (F4/C4/T2; M2 → I3), R-55 (F4/C4/T2; M2 → I3), R-43 (F3/C4/T1; M3 → I3), R-101 (F3/C4/T1; M3 → I3), R-32 (F2/C2/T3; M2→M3 → I3), R-46 (F3/C4/T2; M3 → I3) are analyzed here from a different perspective, with a focus on reflection, verification, and selection.

In literature selection, PhD students combine filtering heuristics with subsequent verification. R-32 describes transparent rules for narrowing the corpus using Elicit (years, design, full-text availability) as a step toward I1–I2, while R-46 documents critical reading and verification of "generated" lists in external databases (Google Scholar), which filters out erroneous references (M3 → I3).

In summary, reflection manifests not in cautious rhetoric but in observable procedures: identifying weaknesses, revising argument structures, adjusting theoretical foundations, transparently selecting sources, and verifying attributions. This aligns with our model (I1–I3) and mechanisms M2–M3: the tool serves as a prompt for analysis, not as a guarantee of correctness.

These findings align with international observations: as emphasized by Lesh and Lancaster [10], doctoral students perceive generative AI as a productive yet potentially risky tool. The authors propose formalized protocols for the ethical use of AI in doctoral education, particularly concerning source verification, plagiarism prevention, and the mitigation of factual distortions. Thus, there emerges a need for ethically oriented digital literacy, one that encompasses critical, autonomous, and responsible use of digital tools.

These findings are consistent with the view that digital tools may increase the cognitive complexity of analysis [6, 7]. These tools enable the visualization of thematic structures, the systematization of sources, and the identification of logical and semantic relationships between ideas, thereby fostering a reflective research environment. However, the final decisions regarding methods, sources, and terminology remain the sole responsibility of the researcher.

4.3 Modelling research logic through digital interaction

The PhD students' materials suggest a pattern of movement from linear exposition toward modelling the logic of inquiry: constructing maps of the problem space, comparing publications over time, and forming multidimensional structures of concepts and evidence. Three tool–indicator relationships operate in this process: M1 (external representation of knowledge) supports I1–I2, M2 (heuristic generation of options) feeds I1–I2, and M3 (verification of evidence and organization of sources) disciplines I2–I3. Within this set, citation-based literature mapping tools function as M1 instruments for structuring the landscape and hypotheses; AI tools for literature review based on semantic search and LLMs serve as M2 for generating quick options and preliminary organization; bibliographic managers serve as M3 for verification and filtering.

Citation-based literature mapping tools provide external representations that link concepts, citation pathways, and thematic clusters. This supports I1 (refinement of the problem and research questions) and I2 (justification of the selection of sources or graph nodes). In the students' materials, the maps and networks serve as a "framework" for subsequent synthesis rather than as an end goal, which aligns with our model (Fig. 1).

Heuristic assistants facilitate the preliminary modeling of argument structure. For example, R-83 describes using Jenni.ai to organize the "Discussion" as a sequence of steps (interpretation → comparison with literature → explanation of discrepancies → limitations → conclusions), supporting I1→I2. R-31 illustrates how Elicit transforms an unstructured query into a transparent selection procedure (filters by year, study design, full-text availability), creating a "framework" for the literature review and reinforcing I2.

Box 05 Examples of Modeling Argument Logic.

> R-83 (F4/C2/T2; M2 → I1/I2): "*Having a set of scattered notes and ideas, I used Jenni.ai to structure the 'Discussion' section. I inserted my key points, and the AI suggested a logical sequence: 1) interpretation of results, 2) comparison with literature, 3) explanation of possible discrepancies, 4) limitations, 5) conclusions and future research. I adjusted this structure, which significantly accelerated the writing process.*"
>
> R-31 (F2/C2/T3; M2→M3 → I1/I2): "*Elicit returned 83 articles, which was too many. Using filters, I narrowed the list to publications from the last two years related to STEM education, specifically review articles with open access. This left a list of 18 key findings that I was able to read. This saved me a lot of time.*"
>
> R-11 (F3/C3/T3; M3 → I2/partly I3): "*Scite helped me justify the choice of methodology for my experiment. I entered a key question about the effectiveness of method X, and the tool provided a summary of studies where this method was successfully applied in similar research contexts. This allowed me to state in the 'Methodology' section that the chosen approach has empirically supported effectiveness, referencing a series of relevant studies found with the help of AI.*"

Verification tools prompt the refinement of the logical foundations of the model. R-11 illustrates the use of Scite to justify methodological choices based on synthesized evidence, directly supporting I2 and, in some cases, extending to I3 (reflective evaluation). This "model-to-evidence" movement prevents digital services from being reduced to purely technical assistants.

In summary, digital interaction in the students' materials is not limited to search activities but involves modeling: from constructing external representations (M1), through heuristic structuring (M2), to the disciplined verification and organization of evidence (M3). This sequence is associated with stronger evidence of problem formulation, procedural selection, and reflective evaluation in the analyzed materials. It aligns with contemporary European approaches to the concept of "second-level digital literacy", the ability not only to operate platforms but to think through them [14, 19].

At the same time, the level of integration of digital tools into research activities varies significantly. As Reyes and Rodríguez [17] note, although doctoral students actively use IT for information search and communication, they often face difficulties in evaluating the reliability of sources, interpreting data, and synthesizing results. These challenges are supported by observations from other researchers. According to them, the effectiveness of digital interaction depends not only on technical skills but also on the ability to critically analyze information flows, methodological flexibility, and the practical relevance of tools within a specific research context [2].

The implementation of digital tools in PhD education is also hindered by time constraints, limited training resources, insufficient preparation of supervisors, and the inertia of academic practices focused on traditional channels of search, publication, and peer review. Despite these challenges, the collected data suggest that in a supportive educational environment, digital platforms may help doctoral students move from linear reasoning toward more multidirectional analysis. They can enhance the level of research structuring and promote the formation of an individual logic for knowledge construction.

4.4 Problems and challenges

Alongside constructive patterns, the students' materials also show inertial practices in interacting with digital services. Typical examples include superficial use of automated functions without connection to research goals (e.g., importing bibliographies without checking formats and sources), accepting AI-generated summaries without editing, and relying on system recommendations without further verification. Such actions are marked by combinations of F1/F2 with C1 and minimal traces of instrumental codes, which do not translate into I2–I3 (methodological justification, reflective evaluation).

A key challenge is the gap between the tool and the research objective. In materials lacking an explanatory transition "tool → procedure → evidence," the tools remain "useful gadgets" rather than elements of evidential logic. In contrast, in reflective texts, students confronted with AI errors or ambiguities demonstrate the ability to correct their trajectory: they verify attributions, filter out incorrect sources, and revise the structure of their arguments.

Box 06 Problematic areas and strategies for resolution.

R-46 (F3/C4/T2; M3 → I3): "*…verified the existence of articles in Scholar… filtered out false sources*" – response to incorrect references in a generated list.

R-32 (F2/C2/T3; M2→M3 → I1/I2): "*…Elicit… 120 articles… applied filters… 25 key findings*" – transition from a large set of references to transparent selection criteria.

R-21 (F4/C4/T2; M2 → I3): "*…used ChatGPT to justify the scope… added two alternative viewpoints in the 'Limitations' section*" – reflexive delineation of AI application boundaries.

R-43 (F3/C4/T1; M3 → I3): "*…ResearchRabbit showed 15 opposing citations… critically reassessed its reliability*" – adjustment of theoretical foundations following verification.

Second challenge – just-in-time self-learning without integration into a logic and analytical workflow. When interaction with tools is limited to technical operations, there is no basis for I1–I2 (problem clarification, justification of procedures). In contrast, a purposeful combination of M1 (external representation of knowledge), M2 (heuristic generation), and M3 (evidence verification) can turn digital actions into elements of a coherent evidentiary sequence.

Third challenge – limited resources and mentorship. When time, institutional support, and formalized verification practices are lacking, the risk of entrenched superficial patterns increases. This is why courses that combine digital tools with methodological rubrics and requirements for justification perform not merely a supportive but a system-building function: they compel each digital step to be explicitly linked to indicators I1–I3.

This is supported by international findings. Doctoral students generally learn digital tools independently, relying on just-in-time learning rather than systematic courses [1]. Even when basic skills are present, many candidates lack confidence in critically applying digital technologies for planning and conducting scientific research.

At the same time, findings indicate that formal training in digital technologies positively impacts the development of methodological culture. Such courses not only improve information literacy but are also associated with the ability to analyze data, justify tool selection, and relate them to the stages of scientific inquiry [8, 11]. Oliinyk et al. [13] emphasize that digital services play an important role in shaping autonomous researchers, those capable of self-directed learning, adapting to innovations, and building their own scientific strategies.

At the same time, as Dowling and Wilson [2] emphasize, the level of integration of digital technologies in PhD education is limited by several persistent factors: lack of time, lack of formalized practices among supervisors, and weak institutional support. These barriers hinder the full use of digital solutions, especially when doctoral students lack support from programs or specialized courses.

4.5 PhD students' practices in using digital tools: survey results

The survey of PhD students complements the qualitative case analysis and reveals typical practices of digital interaction, perceived ease of use, and challenges in applying the tools during actual research work.

The survey illustrated that most participants had 1–3 years of experience (77%), 3–5 years (15%), and over 5 years (8%). Social sciences and humanities dominated (62%). During the literature search, Google Scholar was used most frequently (85%), followed

by Scopus (38%) and Web of Science (31%). In terms of perceived effectiveness, Scopus scored higher (M = 4.3/5), while Google Scholar received a lower average rating (M = 3.8/5) due to filtering difficulties. For the source organization, Zotero was most popular (31%). In data analysis, Excel remained prevalent (67%), whereas Python/R was rated as flexible but with a high entry barrier (8%). Collaborative platforms such as Google Docs (54%) saved time but raised issues with version control and formatting. Presentations were primarily created in PowerPoint (87%), though some participants noted its limitations for academic tasks.

The survey also provided examples aligned with indicators I1–I3.

Box 07 Excerpts from open-ended responses aligned with indicators I1–I3.

R-65 (Elicit; M2 → I1): "Accessibility" → enabled a quick start for literature search with subsequent narrowing of scope.

R-76 (Elicit; M2→M3 → I1/I2): "Speed" → applied filters by year and study type to focus the research field.

R-8 (ChatGPT; M2 → I1/I2): "Convenience and simplicity" → provided alternative formulations, followed by manual verification.

R-70 (ChatGPT; M2 → I3): "Accessibility and speed of draft generation" → subsequently verified against full texts.

R-44 (ResearchRabbit; M3 → I3): "Simplifies work and analysis" → verification of claim support.

R-91 (ResearchRabbit; M3 → I3): "Saves time on verification" → focus on the quality of evidence.

R-22 (Zotero + AI; M3 → I2): "Organization of sources" → appeared to support traceability of citations.

R-82 (Zotero + AI; M3 → I2/I3): "Note-taking speed" → enabled more precise annotations for later verification.

Table 3 presents survey results by aggregating the frequency of use and average ratings of typical statements (e.g., for Google Scholar "simple but hard to filter"). This aligns with the topic of "filtering difficulties" identified in the literature corpus (M3 → I3).

Table 4 presents the proportion of respondents who mentioned AI tools in open-ended responses. Specifically, 56% mentioned LLMs, 33% referred to AI tools for literature review based on semantic search, 22% noted Zotero + AI, and 7% cited citation-based literature mapping tools ($n = 104$). The table also includes brief illustrative quotes related to these tools.

This aligns with empirical patterns:

AI tools for literature review based on semantic search support corpus narrowing and transparent selection rules (M2→M3 → I1/I2);

LLMs function as a heuristic and a trigger for reflective revisions (M2 → I3);

citation-based literature mapping tools enforce verification of claims (M3 → I3);

Zotero (with AI suggestions) structures sources and notes (M3 → I2/I3).

The survey indicates not only which services are popular but also where they may support the indicators of the research reasoning model. The high mention of Elicit and ChatGPT can be explained by their role in question formulation and preliminary structuring (I1–I2). In contrast, ResearchRabbit appears more often in contexts related to

Table 3 Use of digital tools at different stages of research

Research aspect	Popular Tools	% of users	Average effectiveness (1–5)	Typical statement	F-code	C-code	Tool code (Ti)	Mechanism (M)	Indicators (I)
Literature Search	Google Scholar, Scopus, WoS	85 / 38 / 31	3.8 / 4.3 / –	"Simple but hard to filter"	F1	C2	–	M2→M3*	I1, I2
Source & Note Organization	Zotero, Mendeley	31 / 10	4.2 / –	"Helps track sources automatically"	F4	C2	T4 (Zotero/Mendeley)	M3	I2, I3
Data Analysis	Excel, R, Python	67 / 8 / 8	4.0 / 4.5 / 4.5	"R is flexible but complex"	F4	C2→C3	–	M1**	I2
Collaborative Work / Writing	Google Docs, Overleaf	54 / –	–	"Saves time; versioning issues"	F5	C2	–	M1	I2
Presentation of Results	PowerPoint, Canva, LaTeX	87 / 33 / 6	–	"PP is convenient; templates feel generic"	F5	C2	–	M1	I2

*M2→M3: heuristic search followed by subsequent verification/filtering

**For data analysis, the "mechanism" here reflects external structuring/representation (M1); if quality control procedures are applied, M3 is also indicated

Table 4 Mentions of AI tools in open-ended responses ($n = 104$) and aligned qualitative themes with illustrative quotes

Tool	Mentioned, *n* (%)	Leading qualitative theme (theme code)	Illustrative quote (ID)	F-code	C-code	Tool code (Ti)	Mechanism (M)	Indicators (I)
AI tools for literature review based on semantic search	34 (33%)	Search strategies for review (T3)	"Accessibility,""Speed" (R-65, R-76)	F2	C2	T3	M2→M3	I1, I2
LLM	58 (56%)	Quick drafts and idea generation (T2)	"Convenient and simple,""Fast draft" (R-8, R-70)	F4	C2→C4	T2	M2	I1, I2→I3
Citation-based literature mapping tools	7 (7%)	Critical reading, verification of claims (T3)	"Simplifies analysis/ verification" (R-44, R-91)	F3	C3→C4	T1	M3	I3
Zotero with AI suggestions	23 (22%)	Organization of sources and notes (T4)	"Source organization""Fast note-taking" (R-22, R-82)	F4	C2→C3	T4	M3	I2, I3

verification and organization of evidence (I2–I3). Thus, the combination of 3–5 tools, as indicated by respondents, does not reflect mere fragmentation. Instead, it mirrors task distribution across mechanisms M1–M3 throughout the full research cycle. At the same time, some respondents describe barriers, such as the lack of an "all-in-one" solution, system conflicts, and difficulties with English-language queries. These observations support the need for pedagogical guidance and the standardization of minimal verification procedures.

4.6 Educators' observations: external evaluation of doctoral students' digital interaction

External observations were conducted over two academic years in the form of field notes referred to here as educator memos (EM).

The focus was on: types of tool usage, level of reflection, difficulties, group interaction, and initiative. For interpreting the observations, a rubric of levels was used.

L0 (procedural minimum): isolated technical actions without connection to the research goal. Codes were typically F1/F2 + C1. Mechanisms did not extend to I1–I3.

L1 (tool as "convenience"): repeated patterned operations (bibliography import, autofill) with fragmentary explanations. Codes F2/F5 + C2. Sometimes M1, rarely M3. Early forms of I2 emerged.

L2 (modeling): external structuring, heuristics, and transparent selection rules. Codes F2–F4 + C2–C3. M1–M2 were stable, and M3 appeared. Support for I1–I2 was evident.

L3 (reflective integration): evidence verification, framework revision, and delineation of AI boundaries. Codes F3–F4 + C4. M3 was applied systematically. Indicators I2–I3 were realized.

In the observation journals, most cases fell into L1–L2. L3 occurred sporadically but was methodologically significant.

Typical patterns were identified, consistent with the I1–I3 / M1–M3 model.

Pattern A: "Tool without purpose" (L0–L1). Superficial actions such as copying bibliographies or unreflective acceptance of generated abstracts occurred without the transition "tool → procedure → evidence." Codes: F1/F2 + C1. Mechanisms M1/M2 did not transform into M3. Indicators I1–I3 were not indicated.

Pattern B: "Modeling through external representations" (L2). Publication landscape, timelines, and network graphs were used as frameworks for reviews and question formulation. Codes: F4 + C2–C3. Mechanisms M1 and M2 were active. Indicators I1–I2 were engaged.

Pattern C: "Reflective evidence verification" (L3). Verification of counter-claims, revision of theoretical foundations, and clarification of "constraints" occurred after interaction with LLMs. Codes: F3/F4 + C4. M3 was applied consistently. Indicators I2–I3 were realized.

The analysis of observation journals revealed three groups of barriers that systematically hinder PhD students' transition from procedural use of digital services to reflective integration. First, there are technical difficulties: unstable access to resources, format incompatibilities, losses during conversion, and synchronization failures in bibliographies. Under these conditions, digital actions mostly remain at L0–L1. Students perform useful but fragmented operations that are not connected to the logic of evidence. An effective response involves minimal standards for data exchange, such as agreed file formats, naming conventions, and storage chains. Short instructions for verifying typical

steps are also helpful. These measures reduce transactional costs and create space for transitioning to M1–M3 as mechanisms for knowledge construction and verification.

Box 08 Illustrative Episodes from Observations.

EM-01 (L0, F2/C1; no M3): "*The PhD student received from ChatGPT a list of sources for a literature review in the format 'Author (Year). Title.' They copied it into their paper without checking the availability of the articles in open access, their relevance to the topic, or converting them into a consistent bibliographic style (APA). As a result, 3 out of 7 sources were either inaccessible or available only via paid access.*"

EM-02 (L2, F4/C3; M1–M2 → I1–I2): "*While exploring a new topic, the PhD student uploaded a key paper to Connected Papers. Based on the generated network graph, they identified two clear clusters: (1) findings supporting the main hypothesis, and (2) findings offering alternative explanations. This informed the structure of their own literature review, where they compared these two directions.*"

EM-03 (L3, F3/C4; M3 → I3): "*During manuscript preparation, the author used ResearchRabbit to analyze one of the key theories. The tool identified five 'counter-claims' that highlighted methodological considerations in the original study. After reflection, the author did not remove the references but revised the 'Theoretical Background' section, adding a paragraph on the scholarly debate surrounding this theory and its limitations, thereby enhancing the research integrity of the work.*"

EM-04 (L2→L3, F4/C4; M2→M3 → I2–I3): "*After writing the conclusions, the PhD student uploaded the text to Claude with the prompt: 'Analyze the argumentation and suggest potential counterarguments.' The LLM proposed two alternative interpretations of the results that the author had initially overlooked. This prompted a reassessment of their position. As a result, a 'Research Limitations' section was added, presenting these alternative explanations and justifying why they were less likely. This demonstrated the depth of critical analysis.*"

The second group of barriers is conceptual in nature. Some PhD students do not understand how a tool supports the research task. As a result, the chain "tool → procedure → evidence → conclusion" breaks down into isolated actions. Overcoming this barrier is facilitated by explicitly linking each digital step to the indicators of the I1–I3 model. In practice, this is implemented through: for I1, clear formulation of the problem and research questions is required; for I2, justified selection of procedures and source evaluation criteria; for I3, demonstration of reliability checks and the limits of the applied approach. Additionally, it is useful to require PhD students to provide a brief methodological commentary for each platform used. They should explain the function of the tool, its limitations, and how the results were verified.

The third group concerns reflective barriers. PhD students often accept generated texts, source lists, or summaries without sufficient critical revision. Moving these practices to the L3 level requires formalized reflective procedures. An effective measure is the mandatory micro-section "What could be wrong?" in each interim report. Verification of counter-claims for key sources using citation-based literature mapping tools or manual screening is also crucial. Reliability is further enhanced by combining negotiated agreements among analysts with documentation of decisions in a decision log, peer debriefing in small groups, and targeted member checking of selected interpretations.

Together, these procedures may shift digital interaction from mere technical convenience to intellectual responsibility. M3 begins to systematically support I2–I3. Digital services cease to be "black boxes" and become tools for the construction and verification of evidence.

Observations of small group work revealed varying levels of PhD student engagement in collective use of digital tools. At least three stable interaction configurations were identified, each correlating differently with mechanisms M1–M3 and indicators I1–I3.

The "instructional imitation" model is characterized by reproductive replication of steps demonstrated by an educator or peer. Discussions are limited to technical actions (bibliography import, basic search), with dominant codes F1–F2 and C1–C2. Tools function as conveniences rather than elements of evidence-based reasoning. Mechanism M1 appears episodically, M3 is practically inactive, and indicators I1–I3 remain indistinct.

The "collaborative modeling" model is associated with the construction of external knowledge representations (citation networks, concept maps, timelines) and their collegial discussion. Groups link graph nodes to research questions, define criteria for source inclusion, and record interim versions of the review. Here, F4 and C2–C3 are strengthened, M1 operates systematically, M2 appears periodically, and indicators I1–I2 acquire operational significance.

The "reflective integration" model involves collective verification of assumptions, tracking counter-claims for key papers, clarifying the limits of LLM use, and recording decisions in shared journals. Codes F3–F4 emerge with emphasis on C4. Mechanism M3 becomes dominant. Indicator I3 manifests through justified changes in the argument structure and source selection criteria.

Initiative within groups correlates with organizational conditions. The following practices appear to play a notable role:

(a) Mandatory comparison of tools based on their role in the chain "tool → procedure → evidence";
(b) Brief pitching of decisions with public justification of choices, which supports I1, I2, and I3;
(c) Joint "evidence audits," including verification of counter-claims, tracking data provenance, and maintaining a decision log.

These practices reduce dependence on instructions and shift interaction from the level of procedural exchange to collective construction of evidence.

At the same time, situations of reduced participation were observed: formal presence without contribution to analysis, confinement to a "personal" topic, and avoidance of comparing alternative tools. In such cases, it is advisable to implement structured roles (facilitator, skeptic, verifier, scribe) and minimal process materials (shared decision matrix, inclusion/exclusion criteria, template for "negative cases"). These interventions strengthen M3, make steps toward I2–I3 visible, and may foster initiative not through individual "activity," but through responsibility for elements of collective evidence.

Thus, group interaction promotes the transition from procedural task execution to reflective integration of digital tools when collaborative work is organized around evidence rather than mutual assistance in using services. Under these conditions, mechanisms M1–M3 sequentially support indicators I1–I3, and initiative takes the form of responsibly justified actions within the collective research logic.

5 Conclusion and implications

The results suggest that the integration of digital tools with methodological training can support research reasoning processes among PhD students. Within the course "Modern IT in Scientific Research," the analyzed materials contained patterns consistent with movement from procedural work with services toward methodologically guided actions, where tools functioned as cognitive mediators. PhD students' materials provided evidence aligned with the indicators of the research reasoning model: clarifying the research problem and questions (I1), justifying the selection of procedures and source evaluation criteria (I2), and reflectively assessing evidence reliability and the limits of AI application (I3). These patterns can be interpreted through three mechanisms of digital interaction: external knowledge representation and landscape structuring (M1), heuristic generation of options and preliminary modelling (M2), and evidence verification and source organization (M3). Alignment of these observations with data excerpts, tool mentions in open responses, and educator observation journals suggests that digital tools can become part of argument construction rather than merely convenient add-ons when they are embedded in methodologically guided tasks.

The theoretical contribution lies in the operationalization of the construct "research reasoning" through the three indicators I1–I3 and their linkage to mechanisms M1–M3. Instead of a descriptive list of competencies, observable patterns are presented: construction of external representations to clarify the problem, use of heuristics as triggers for structuring, and systematic verification as a discipline of evidential reasoning. Thus, the proposed framework allows tracking not mere "tool mastery" but the logic of transitions between analytical states of reasoning during actual research activity.

The practical implications concern the organization of graduate education. First, mastery of digital tools should not occur in isolation but within research tasks with an explicit link "tool → procedure → evidence." Second, implementing minimal verification standards is advisable: checking counter-claims for key sources, transparent inclusion and exclusion criteria, brief decision logs, and negative case analyses. Third, assignments requiring reflective micro-sections in reports ("What could be wrong?"; "Limits of the applied approach") and public pitching of tool choices justified according to I1–I3 appear productive. Fourth, group work is more effective when organized as collaboration around evidence: distributed verification roles, brief evidence audits, and comparison of tools according to their role in the model.

Implications for institutional policy include the introduction of interdisciplinary modules on digital methodology. Institutions should also train educators to work with AI tools as instruments for constructing and verifying evidence. Another important step is the creation of research support laboratories and the implementation of ethical protocols for AI use. Together, these measures may help cultivate researchers who think through modelling, verify hypotheses across multiple sources, and can justify the limits of applying digital services.

Limitations. The empirical material covers a corpus of student materials from two Ukrainian universities and is qualitative in nature. Therefore, generalization to other programs should be made with caution. We did not calculate theme frequencies or intercoder agreement coefficients, prioritizing reflexive thematic analysis, thick description, and analytical generalization. Any numerical references are provided only for transparency of data sources and are not interpreted as effects. The results may also have been

influenced by differences in students' levels of digital preparedness and the thematic focus of their research projects.

Future studies would benefit from mixed-methods designs with independent pre- and post-course measurements, comparative analyses of programs with different models of integrating M1–M3, and quantitative validation of the proposed indicators I1–I3.

In sum, integrating digital technologies into doctoral methodological training may foster a research culture in which technical proficiency is combined with evidential rigor and reflective responsibility. This creates opportunities for developing academic autonomy and intellectual mobility among PhD students and provides a foundation for future empirical testing of the proposed model.

Appendix A

Questionnaire

1. General Questions

How many years have you been engaged in scientific research?
What is your scientific specialty?

2. Questions About Research Stages and Digital Tools

Literature Search

Which digital tools (databases, platforms) do you use to search for scientific sources? (e.g., Google Scholar, PubMed, Scopus, Web of Science)
How do you rate the effectiveness of these tools on a scale from 1 to 5?

Data Organization and Note-taking

Do you use specialized software for managing literature (Zotero, Mendeley, EndNote)? How often?
What advantages and disadvantages of these tools can you identify?

Data Analysis

What software do you use for statistical analysis/ data visualization? (R, Python, SPSS, Excel, etc.)
Are you satisfied with their functionality? Why?

Writing and Collaboration

Which tools do you use for collaborative work on articles? (Overleaf, LaTeX, Google Docs, Trello)
Do they help save time?

Presentation of Results

Which software do you use to create presentations?
Do you find them convenient for academic purposes?

3. Open-ended Questions for In-depth Analysis

Which digital tools do you consider most useful in your scientific work? Please explain why.
What difficulties do you face when using digital tools?
What would you like to change or improve in the existing tools?

Appendix B

Observation Sheet for Doctoral Students
Course: Modern IT in Scientific Research
Educator Name: _____
Date of Session: _____
Module / Topic: _____
Number of Doctoral Students Present: _____
Session Format: _____

1. Type of Digital Tool Use (check all observed).
☐ Use tools only according to instructions (procedural reproduction)
☐ Try to understand how the tool works (functionality analysis)
☐ Show interest in potential research applications
☐ Independently modify/combine tools for their tasks
☐ Justify the choice of digital tools from methodological perspectives
Comments / Examples of Thinking in Action:_____

2. Level of Reflection on Research Process and Use of Digital Tools.
☐ Ask about limitations/reliability of the tool
☐ Critically analyze results obtained using digital tools (verification)
☐ Discuss the impact of digital tools on research conclusions
☐ Identify possible biases, errors, and risks
☐ Superficial acceptance of results without attempts at interpretation
Examples of critical or uncritical behavior:_____

3. Modeling Research Logic through Digital Interaction.
☐ Formulate research questions/hypotheses based on digital analytics
☐ Use digital tools to structure data/literature/arguments
☐ Interpret digital data in light of theoretical approaches
☐ Only perform technical steps without attempting to build research logic
Description of examples (individual or group):_____

4. Difficulties Encountered by Doctoral Students.
☐ Technical (access, errors, compatibility)
☐ Conceptual (do not understand how the tool supports research)
☐ Logical (disrupted sequence of steps or understanding of connections)
☐ Reflective (do not question the appropriateness/limitations of digital tools)
☐ Time/organizational
☐ Other (specify): _____
Description of situations / examples of behavior:_____

5. Group Interaction as a Factor in Developing Research Reasoning.
☐ Doctoral students actively discuss ways to use digital tools
☐ Exchange views on result interpretation
☐ Collaborate on building the logic of analysis or review
☐ Passivity or isolation observed
☐ Competition or dominance of certain participants
Comments / situations illustrating interaction style:_____

6. Interest in the Topic and Initiative in Learning.
☐ Propose their own examples/tools
☐ Ask questions beyond the assigned tasks
☐ Complete tasks without initiative
☐ Avoid active participation
Explanation / examples of active or passive participation:_____

7. Identified Potentials for Further Course Development / Doctoral Student Support.
_____

8. Additional Observations (behavior of individual students, unexpected difficulties, successful cases, etc.).
_____

Author contributions
All authors were involved in data collection and analysis. While all authors contributed to the preparation of the manuscript, DM was responsible for writing the Introduction and Literature Review. MS and KS contributed to writing

the Discussion section. SO authored the Methodology, Results, and Conclusions sections. The corresponding author confirms that the manuscript has been read and approved by all listed authors. There are no individuals who meet the criteria for authorship but are not included in the author list. The order of authorship, as indicated on the title page, has been agreed upon by all authors. The corresponding author is the sole point of contact during the editorial process. They are responsible for communication with the other authors regarding progress, editorial revisions, and final approval of the proof. All authors have read and approved the final version of the manuscript. They agree to its submission to Discover Education.

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Data availability
The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
The study was approved by the Ethics Committee of Sumy State Pedagogical University named after A. S. Makarenko. It complied with ethical standards and the provisions of the Ethical Guidelines for Educational Research (European Educational Research Association, 2011). Informed consent was obtained from all individual participants included in the study.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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